Cluster map: urban vulnerabilities and design decisions

Giovanni, Elena and David explore the shared topic through their research. Use the buttons to follow one person around the ring.

KNOWLEDGE

- G Does species choice trade heat against runoff?
- E Which people can synthetic participants stand in for?
- D How can thermal comfort and daylight improve in existing homes?
- ALL *What gets lost in translation?*

USERS

- G Computational designer; municipal climate planner; urban hydrologist
- E Planners; sensory-sensitive and neurodivergent people
- D Owners and management; advising architect; residents consulted
- ALL *Designers who are not climate experts*

IMPACTS

- G Heat stress (mean UTCI, 12–17 h on a hot design day) and storm runoff on Porta streets
- E Whose experience counts; risk of tokenism
- D Hypothesis: better thermal comfort and daylight; trade-offs need checking
- ALL *Decisions affecting life in hotter summers*

VERIFICATION

- G Interception vs field data; current trees as baseline; Florence transfer
- E Each persona vs its own participant; bias gap
- D Measurements, interviews and simulations; technical review to agree
- ALL *Who checks that the translation is right?*

DRIVERS

- G Plane trees to be cut toward the 15% species cap; pluvial floods
- E Planners adopting AI synthetic participants
- D Improve current conditions, keeping site layout and openings fixed
- ALL *Hotter summers*

CLUSTER TOPIC

Urban vulnerabilities and design decisions

How computation turns information about places, and the people in them, into options designers can understand and question.

SHARED DISCIPLINE: COMPUTATIONAL DESIGN

Giovanni

UNIT one plane-tree position in Porta, Barcelona
DISCIPLINES urban hydrology and climatology, landscape, plant ecology

Elena

UNIT one photo of a Vienna public space
DISCIPLINES participatory planning, AI, sensory experience

David

UNIT an intervention in existing homes at Ribera Payandé
DISCIPLINES thermal comfort, daylight, building retrofit

APPLICATIONS

- G Species and position for each plane tree replaced
- E Persona ratings of places and of proposed changes
- D Shading, vegetation, roofs and finishes; site layout and openings fixed
- ALL *Comparing options before they are built*

BY-PRODUCTS

- G Porta trait table; LiDAR tree table; scenario results
- E Rated photos of five Vienna places; corpus of persona reasons
- D Home profiles, field records and comparisons of interventions
- ALL *Reusable data; permissions and openness to agree*

DATA IN

- G ICGC LiDAR 2021, tree inventory, PDISBA storms, weather, traits
- E Photos of five Vienna places; survey profiles and lived experience
- D Drawings, materials, local weather, measurements and resident interviews
- ALL *Who or what is missing from the data?*

SERVING INFRASTRUCTURE

- G Python core, Grasshopper and Ladybug; Infrared.city to confirm
- E Paid model API (Claude Haiku 4.5, zero data retention); Google Forms survey
- D Rhino/Grasshopper, Python, Ladybug, Infrared.city and ClimateStudio; to assess
- ALL *Lock-in: own code and paid services*

DATA OUT

- G Ranked layouts; shade and runoff maps
- E Mean score and full spread, with persona reasons, on a map
- D Diagnosis and options: thermal and daylight effects, assumptions and limits
- ALL *An answer the designer can question*

MONITORING

- G Open: no tracking after planting yet
- E Blind checks keep testing personas against real people
- D Record simulated effects; field follow-up only if changes are built
- ALL *Same metrics for before-and-after comparisons*

Giovanni T1 Shaamala et al. 2025 T2 Mannucci et al. 2025 P1 SylvCiT, Nicol et al. 2026 P2 Davey Tree Benefits Engine P3 GreenInCities

Elena T1 Arnstein 1969 T2 Argyle et al. 2023 P1 MacLennan et al. 2023 P2 Raymond et al. 2025 P3 Yao et al. 2026

David T1 Huang et al. 2025 T2 Cheung et al. 2025 P1 Jia, Lu & Yin 2023 P2 Taghavifard & Cheraghi 2026 P3 Sadeghipour Roudsari & Pak 2013

T = theory papers, P = practice papers.

Based on the group's draft and the faculty's cluster-map template. All three proposals are awaiting discussion.